effect. However, it was very unclear to me in 1985 whether a wormhole can contain *enough* exotic matter to hold it open. So I did two things.

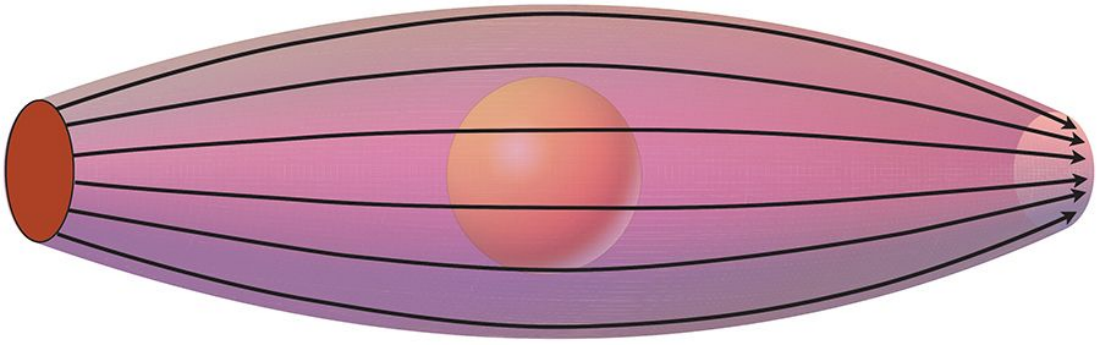


Fig. 14.5. The Sun or a black hole bends a beam of light inward.

First, I wrote a letter to my friend Carl suggesting that he send Eleanor Arroway to Vega through a wormhole rather than a black hole, and I enclosed a copy of the calculations by which I had shown that the wormhole must be threaded by exotic matter. Carl embraced my suggestion (and wrote about my equations in the acknowledgment of his novel). And that is how wormholes entered modern science fiction—novels, films, and television.

Second, with two of my students, Mark Morris and Ulvi Yurtsever, I published two technical articles about traversable wormholes. In our articles, we challenged our physicist colleagues to figure out whether the combined quantum laws and relativistic laws permit a very advanced civilization to collect enough exotic matter inside a wormhole to hold it open. This triggered a lot of research by a lot of physicists; but today, nearly thirty years later, the answer is still unknown. The preponderance of the evidence suggests that the answer may be NO, so traversable wormholes are impossible. But we are still far from a final answer. For details, check out *Time Travel and Warp Drives* by my physicist colleagues Allen Everett and Thomas Roman (Everett and Roman 2012).